# Developer Data Specification & Integration Guide

Authoritative Data Architecture, Synthetic JSON Datasets, Clinical Invariants & KPI Computation Rules

**Executive Purpose & Scope:** This document provides the engineering handover specification for developers building or maintaining the **STAND-HF Clinical Dashboard** independently from the primary registry eCRF. It defines the canonical JSON data models, validation invariants, clinical lookups, and algorithmic formulas governing all dashboard telemetry (4/4 GDMT optimization, follow-up window adherence, and functional outcomes). The companion datasets in srs/data/ provide full realistic test coverage with zero external backend dependencies.

## 1. Data Package Inventory & Architecture

The dataset assets are organized under srs/data/ in pure JSON and vanilla scripts:

| Asset Path | Type | Scope / Volume | Technical & Clinical Role |
|---|---|---|---|
| `srs/data/dummy_data.json` | JSON | **44 Patients**<br>132 Visit Slots | Master registry dataset matching browser localStorage (standhf_ecrf_v1). Contains patient records, visits (Baseline, 3M, 6M), Section B vitals, Section C GDMT pillars, Section D adverse events, audit history, and file attachments. |
| `srs/data/access.json` | JSON | 4 Users · 4 Sites<br>3 Geo Zones | System RBAC model. Configures Super Admin, Central Team, Site Investigators, permitted hospital sites (RIC, LHR, KHI, ISB), and zones (North, Central, South). |
| `srs/data/lookups.json` | JSON | Complete Master Lists | Clinical dropdown dictionaries: GDMT generic drugs, allowed strengths, default frequencies (OD, BD, TDS), NYHA classes, and non-initiation reasons. |
| `srs/data/README.md` | Doc | Developer Manual | Full Markdown integration guide detailing TypeScript interfaces, API schemas, and calculation steps. |

## 2. Quickstart Integration Patterns

- **Frontend Frameworks (React, Vue, Vite, Next.js):** Import directly as a local JSON module: import dataset from './data/dummy_data.json'; const patients = dataset.pts;
- **Mock REST API Server:** Launch via standard tooling: npx -y json-server --watch srs/data/dummy_data.json --port 4000 to serve endpoints GET /pts and GET /pts/:id.
- **Browser LocalStorage Injection:** Run localStorage.setItem('standhf_ecrf_v1', JSON.stringify(dataset));. Both dashboard/index.html and records/index.html will immediately hydrate.

## 3. Access Control & Mandatory Site-Scoping Invariant

**Data Isolation Rule:** Users with role investigator (e.g. emily.carter at RIC/LHR or adnan.khan at KHI/ISB) **MUST NEVER** view, aggregate, or edit records outside their assigned site list (mySites()). Users with role central or superadmin possess global visibility across all 4 medical centres.

## 4. Patient Entity & Demographics Schema

Each patient object in the pts array represents an enrolled registry subject:

| Field Path | Type | Format / Allowed Values | Clinical Rule & Validation Behavior |
|---|---|---|---|
| `id` | string | {SITE}-{001} | Unique study identifier (e.g. RIC-001). Auto-sequenced by site. |
| `site` | string | RIC, LHR, KHI, ISB | Hospital code: Rawalpindi, Lahore, Karachi, or Islamabad. |
| `inv` | string | Full Name | Principal Investigator responsible for this subject. |
| `name` | string | 2+ characters | Patient full name. Required for visit completion. |
| `contact` | string | **11 Digits** (0300...) | Primary phone number. Must match /^\d{11}$/. Non-digits stripped. |
| `consent` | date | YYYY-MM-DD | Mandatory date. Cannot be in future. Baseline date defaults to this. |
| `age` | number | 18 to 150 years | Inclusion requires age ≥ 18. Advisory banner triggered if 111–150. |
| `sex` | enum | 'Male' \| 'Female' | Patient biological sex. |
| `etio` | enum | 'Ischemic' \| 'Non-ischemic' \| 'Other' | Heart failure primary etiology. If 'Other', etioO is mandatory. |
| `sync` | enum | 'Synced' \| 'Pending' | Local sync status with central server. Yellow badge when pending. |
| `v` | object | { b, m3, m6 } | Visit dictionary: Baseline (b), 3 Months (m3), and 6 Months (m6). |

## 5. Visit Sections B & C Specification

### Section B: Clinical & Functional Assessment (Single Unified Card)

| Parameter | Key Path | Clinical Bounds | Validation Rules & Form Behavior |
|---|---|---|---|
| Weight | `v[k].wt` | 30.0 – 250.0 kg | Decimal number (step 0.1). Required to lock visit. |
| Heart Rate | `v[k].hr` | 30 – 220 bpm | Resting pulse rate. Soft warning triggered outside range. |
| Blood Pressure | `v[k].sbp, dbp` | SBP 50–260, DBP 30–160 | **Hard error:** SBP must exceed DBP (sbp > dbp). |
| NYHA Class | `v[k].nyha` | I, II, III, IV | Functional classification. Class IV triggers advisory check. |
| LVEF (%) | `v[k].lvef` | 5 – 80% (Inclusion ≤ 40%) | **Baseline & 6M ONLY. Strictly N/A at 3 Months.** |
| 1MSTS Reps | `v[k].msts` | 0 – 60 repetitions | 1-Minute Sit-to-Stand functional capacity test. |
| Stop Reason | `v[k].stop` | Standard enum | Documented if 1MSTS stopped early (Dyspnea, Syncope, etc.). |

### Section C: Foundational GDMT (4 Pillars)

Selecting a generic drug automatically marks Status as 'Active'. Frequency and Strength dropdowns expand to full width on touch tablets.

| Pillar | Pharmacology | Approved Generics | Strengths & Recommended Dosing Frequencies |
|---|---|---|---|
| `arni` | ARNI | Sacubitril Valsartan | 25mg, 50mg, 100mg, 200mg · **BD** |
| `bb` | Beta-blocker | Bisoprolol, Carvedilol, Nebivolol | Bisoprolol: 2.5, 5, 10mg (OD) · Carvedilol: 3.125, 6.25, 12.5, 25mg (BD) · Nebivolol: 1.5, 2.5, 5, 10mg (OD) |
| `mra` | MRA | Spironolactone, Eplerenone, Finerenone | Spironolactone: 12.5, 25, 50mg (OD) · Eplerenone: 25, 50mg (OD) · Finerenone: 10, 20, 40mg (OD) |
| `sg` | SGLT2 inhibitor | Dapagliflozin, Empagliflozin, Sotagliflozin | Dapagliflozin / Empagliflozin: 10mg (OD) · Sotagliflozin: 200, 400mg (OD) |

**Non-Initiation Reasons (v[k].rs):** If any pillar is 'Not started' or 'Stopped', at least one reason must be flagged: hypo (Hypotension), brady (Bradycardia), renal (Renal dysfunction), k (Hyperkalemia), intol (Intolerance), cost (Cost barrier), pref (Patient preference), or phys (Physician decision), plus free-text rsO.
**Section D (Events v[k].ev):** Tracks ev.hf (HF hospitalization), ev.all (All-cause hospitalization), ev.death (Death at 3M/6M), and ev.int (Major intervention description).

## 6. Dashboard KPI & Metric Calculation Algorithms

Dashboard developers must implement these standardized calculation functions to ensure consistency across views:

**Algorithm 1: Quadruple GDMT Optimization Rate (Protocol Target: ≥ 70%)**

```
function act(v) { return ['arni','bb','mra','sg'].filter(q => v.g?.[q]?.status === 'Active').length; }
```

- For each patient, identify their **most recent completed visit** (last = ['m6','m3','b'].find(k => p.v[k]?.done)).
- Evaluate act(p.v[last]). Quadruple therapy requires all 4 pillars to be 'Active'.
- **Dashboard KPI Formula:** Rate (%) = (Count of patients with 4 active pillars / Total patients with ≥1 visit done) × 100

**Algorithm 2: Follow-Up Protocol Window Adherence (On-Time Follow-up Rate)**

Evaluates whether completed follow-up visits occurred within protocol target windows:

- **Month 3 (m3):** Target day 90 ± 14 days (Window: Day 76 to Day 104 from Baseline date).
- **Month 6 (m6):** Target day 180 ± 21 days (Window: Day 159 to Day 201 from Baseline date).

```
function onTime(p, k) { const d = Math.round((day(p.v[k].date) - day(p.v.b.date)) / 864e5); return Math.abs(d - (k==='m3'?90:180)) ≤ (k==='m3'?14:21); }
```

- **Adherence KPI Formula:** OnTimeRate (%) = (Total on-time visits / Total completed follow-up visits) × 100

**Algorithm 3: Primary Functional & Echocardiographic Endpoints (Δ Baseline → 6 Months)**

Evaluated exclusively across the cohort of patients having completed both Baseline and Month 6 visits:

- **Mean Δ 1MSTS Capacity:** MeanDeltaMSTS = Σ (p.v.m6.msts - p.v.b.msts) / Count(completed_cohort) (Repetitions in 60s).
- **Mean Δ LVEF Remodeling:** MeanDeltaLVEF = Σ (p.v.m6.lvef - p.v.b.lvef) / Count(completed_cohort) (Absolute percentage points).

**Algorithm 4: Visit Status Classification State Machine (vstat)**

Drives follow-up queues, upcoming visit calendars, and status pill badges:

1. **'Completed':** v.done === true
2. **'Missed':** Today > BaselineDate + TargetDays + WindowTolerance + 30 days, AND no data has been entered.
3. **'Overdue':** Today > BaselineDate + TargetDays + WindowTolerance, AND visit not completed.
4. **'Due':** Today ≥ BaselineDate + TargetDays - WindowTolerance, AND visit not completed.
5. **'In Progress':** Visit uncompleted but draft data exists (vitals, drugs, or attached file).
6. **'Not Started':** Visit window has not yet opened.

**Algorithm 5: Adverse Clinical Event Telemetry (Section D Aggregations)**

Counts cumulative adverse outcomes reported across follow-up visits (3M and 6M):

- **HF Hospitalizations:** pts.filter(p => ['m3','m6'].some(k => p.v[k]?.ev?.hf)).length
- **All-Cause Hospitalizations:** pts.filter(p => ['m3','m6'].some(k => p.v[k]?.ev?.all)).length
- **Deaths Recorded:** pts.filter(p => ['m3','m6'].some(k => p.v[k]?.ev?.death)).length

## 7. Dummy Dataset Telemetry & Test Coverage

The dataset in srs/data/dummy_data.json contains 44 realistic records verifying all dashboard states:

| Metric / Indicator | Dataset Result | Protocol Target | Dashboard Component Tested |
|---|---|---|---|
| Enrolled Patient Roster | **44 Patients** | 40–50 Subjects | Roster table, patient search, and site distribution. |
| Site Allocation | 11 RIC · 11 LHR · 11 KHI · 11 ISB | Balanced 4 centres | Site league table, zone filter, and investigator scoping. |
| Baseline Completed | 44 of 44 (100%) | 100% enrolled | Enrolment KPI counter and recruitment trajectory. |
| 3 Months Completed | 28 of 44 (63.6%) | Longitudinal follow-up | 3M Milestone completion counter and Phase Dial. |
| 6 Months Completed | 16 of 44 (36.4%) | Full study completion | 6M Milestone completion counter and Phase Dial. |
| Protocol Window Adherence | **100% On-Time** | ≥ 80% Adherence | Clinical performance on-time KPI widget. |
| Quadruple GDMT Coverage | **75%** (33 of 44 patients) | ≥ 70% Target | GDMT coverage bar chart, pill tiles, and donut. |
| Mean Δ 1MSTS Outcome | **+10.5 repetitions** | Functional gain > 0 | Functional outcome card (Δ 1MSTS baseline → 6M). |
| Mean Δ LVEF Remodeling | **+9.9 percentage points** | Echocardiographic gain | Echocardiographic remodeling card (Δ LVEF). |
| Follow-Up Action Queue | 3 Overdue · 7 Due | Queue active > 0 | Tests urgent follow-up queue and overdue alert badges. |
| Adverse Clinical Events | 3 HF Hosp · 1 Death at M6 | Safety reporting | Adverse clinical event telemetry tiles. |

## 8. Sample Patient Record Snippet (JSON)


```
{
  "id": "RIC-001", "site": "RIC", "inv": "Emily Carter", "owner": "emily.carter",
  "name": "Ali Raza", "contact": "03005550100", "consent": "2026-04-06", "age": 38, "sex": "Male", "etio": "Ischemic",
  "sync": "Synced", "files": { "b": [{"id": "f_ric001_b", "name": "prescription-RIC-001-b.png", "type": "image/png"}] },
  "v": {
    "b": {
      "date": "2026-04-06", "done": true, "wt": 65.0, "hr": 76, "sbp": 112, "dbp": 70, "nyha": "II", "lvef": 22, "msts":
8, "stop": "",
      "g": {
        "arni": { "drug": "Sacubitril Valsartan", "strength": "50mg", "freq": "BD", "status": "Active" },
        "bb":   { "drug": "Bisoprolol fumarate", "strength": "2.5mg", "freq": "OD", "status": "Active" },
        "mra":  { "drug": "Spironolactone", "strength": "25mg", "freq": "OD", "status": "Active" },
        "sg":   { "drug": "Dapagliflozin", "strength": "10mg", "freq": "OD", "status": "Active" }
      },
      "ev": { "hf": false, "all": false, "int": "Coronary Angiography" }, "ini": "EC"
    },
    "m3": { "date": "2026-07-03", "done": true, "wt": 63.5, "hr": 72, "sbp": 110, "dbp": 69, "nyha": "I", "msts": 12, ...
 },
    "m6": { "date": "2026-09-30", "done": true, "wt": 62.5, "hr": 68, "sbp": 108, "dbp": 68, "nyha": "I", "lvef": 30, "ms
ts": 17, ... }
  }
}
```


**CERTIFICATION & DEVELOPER RELEASE:**
This document and accompanying datasets in srs/data/ are validated for dashboard independent development.

**STATUS: Approved Technical Release**
**BUILD: STAND-HF Portal v1.0**